

Testing Plan

MPPT Buck Converter — Bring-up and MPPT Demonstration

Overview

Testing proceeds in two phases. Phase A is bench bring-up on a fixed 19 V supply: prove the converter switches and regulates, in a controlled order that isolates faults. Phase B is the MPPT demonstration on the actual panel: compare three ways of connecting the panel to the load and show that active tracking wins.

The one rule that prevents damage: always flash the intended sketch *before* turning on the supply. The ESP32 re-runs whatever is in flash the instant it powers up. Also keep duty ≤ 0.75 on the 19 V bench so the 16 V output cap is never overvolted.

Phase A — Bench bring-up (19 V supply, 10 Ω load)

#	Test	What you do	Pass criteria
1	Gate drive, static	No load. Hold GPIO low, then high. Measure gate, switch node, Q2 collector.	GPIO low: gate 0 V. GPIO high: gate goes strongly negative, Q2 collector near 0 V.
2	PWM reaches gate	1 kHz test PWM, sweep duty 25/50/75%. Read Q2 collector with a DMM (it averages).	Collector voltage tracks (1–duty) linearly across the sweep.
3	Power path / regulation	10 kHz. Load connected. Start duty low, step up, measure Vout at each step.	Vout tracks $D \times V_{in}$. Inductor stays cool. Supply current sane (~ 1 A near top).
4	Thermal soak	Hold the top duty five minutes. Touch Q1, D1, R2, L1.	All warm at most. Gate clamp reads ~ -10 V (zeners now engaged).
5	Instrumentation	I2C scan for both INA226. Cross-check each reading against the DMM.	Both answer (0x40, 0x41). Readings match DMM. Power-in > power-out.

Fault-isolation habit

If a test fails, the informative measurement is almost always **Vout at a known duty with the load connected**, plus **which part is hot**. Resistance readings on a live board full of capacitors and junctions are misleading and should not be trusted for diagnosis. When something reads open that shouldn't, tug the jumper — solid-core wire can look seated while making no contact.

Phase B — MPPT demonstration (panel, outdoors)

Three circuits are compared for how much power actually reaches the 10 Ω load. The point of the experiment is that the panel has one best operating point, and only the tracking converter can hold the panel there while feeding a load that wants something different.

Circuit	Path	Expectation
A — Direct	Panel → INA226 → load	Worst. Load resistance drags the panel off its peak into the current-limited region.
B — Fixed duty	Panel → INA226 → buck (fixed D) → INA226 → load	Ties the tracker in steady sun; falls behind when conditions change.
C — MPPT	Panel → INA226 → buck (P&O) → INA226 → load	Best across changing light. Continuously re-finds the panel's peak.

Why C beats A (the core result)

A buck at duty D makes the panel see $\text{load_resistance} \div D^2$. The panel's ideal load is about $33\ \Omega$ ($18.34\text{ V} \div 0.55\text{ A}$), but the actual load is $10\ \Omega$. Directly connected (A), the $10\ \Omega$ forces the panel down to roughly 6 V and only $\sim 3.5\text{ W}$ is drawn out of a possible 10 W . Through the converter at $D \approx 0.55$ (C), the $10\ \Omega$ is transformed to $\sim 33\ \Omega$, the panel sits at 18 V and delivers its full power, and tracking keeps it there as the sun moves. The bigger the mismatch between load and the panel's ideal, the bigger the MPPT win.

Measurement protocol

- **Metric:** average power delivered to the load (load-side INA226) over a 5-minute window.
- **Sampling:** ESP32 logs power every 100–200 ms and averages in firmware, rather than eyeballing.
- **Schedule:** run all three, three times across the day, to capture different sun angles.
- **Interleave, don't block:** cycle A→B→C in short chunks and note panel open-circuit voltage before each, so drifting irradiance averages out instead of favoring one circuit.
- **Also log tracking efficiency for C:** load power $\div$ panel power available — a stronger result than raw watts because it is less sensitive to how bright the day happens to be.

Fairness notes

- Only one converter drives the load at a time. Physically disconnect the idle one — a converter that is off but still wired can back-feed and skew the reading.
- Fix circuit B's duty once and leave it; tuning it per session turns it into manual MPPT and erases the contrast.
- For circuit A the two INA226 measure the same electrical point, so a single meter suffices there.
- Expect real panel watts below the 10 W rating — August sun in Illinois is well under lab conditions. The ratios between A, B and C are the result, not the absolute numbers.